

CUET PG MCA (DSA PYQS)

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Topics Covered: Data Structures & Algorithms, Asymptotic Analysis, Arrays, Searching & Sorting, Linked Lists, Stacks, Queues, Trees (BST, B/B* Trees), Graph Theory & Algorithms.

Question 1

CUET PG 2021 | Linear Search

What is the worst-case time complexity of linear search algorithm?

- (a) $O(n)$
- (b) $O(1)$
- (c) $O(\log n)$
- (d) $O(n^2)$

Question 2

CUET PG 2021 | Binary Search

Which of the following is not an application of binary search?

- (a) To find the lower/upper bound in an ordered sequence
- (b) Union of intervals
- (c) Debugging
- (d) To search in unordered list

Question 3

CUET PG 2021 | Time Complexity

The complexity of binary search algorithm is:

- (a) $O(n)$
- (b) $O(\log n)$
- (c) $O(n^2)$
- (d) $O(n \log n)$

Question 4

CUET PG 2021 | Iteration Count

Given an array $A = \{55, 66, 77, 88, 99\}$ and a key = 88. How many iterations are done until the element is found?

- (a) 1
- (b) 3
- (c) 4
- (d) 2

Question 5

CUET PG 2021 | Time Complexity

Which of the following operations is not $O(1)$ for an array of sorted data? You may assume that array elements are distinct.

- (a) Find the i -th largest element
- (b) Delete an element
- (c) Find the i -th smallest element
- (d) All of the above

Question 6

CUET PG 2021 | Quick Sort

Quicksort is a:

- (a) Greedy algorithm
- (b) Divide and conquer algorithm
- (c) Dynamic programming algorithm
- (d) Backtracking algorithm

Question 7

CUET PG 2021 | Pseudocode

Informal high-level description of an algorithm in English is called:

- (a) Function
- (b) Class
- (c) Pseudocode
- (d) None of the above

Question 8

CUET PG 2021 | Scope of Variables

What is the scope of the variable declared in the user-defined function?

- (a) In whole program
- (b) Only inside the {} block
- (c) Both 1 and 2
- (d) None of these

Question 9

CUET PG 2021 | Algorithm Analysis

The time factor when determining the efficiency of an algorithm is measured by:

- (a) Counting microseconds
- (b) Counting the number of key operations
- (c) Counting the number of statements
- (d) Counting the kilobytes of algorithm

Question 10

CUET PG 2021 | Flowchart Symbols

The geometrical figure shown in flowchart represents:

- (a) Input
- (b) Connector
- (c) Decision
- (d) Looping

Question 11

CUET PG 2021 | Flowchart

Which of the following is a structured programming technique that graphically represents the detailed steps required to solve a program?

- (a) Algorithm
- (b) Pseudocode
- (c) Flowchart
- (d) Top-down design

Question 12

CUET PG 2021 | Stack

To evaluate an expression without any embedded function calls:

- (a) One stack is enough
- (b) Two stacks are needed
- (c) As many stacks as the height of the expression tree are needed
- (d) A Turing machine is needed in the general case

Question 13

CUET PG 2021 | Linear Data Structures

Which of the following data structures is a linear type?

- (a) Graph
- (b) Tree
- (c) Binary Tree
- (d) Queue

Question 14

CUET PG 2021 | Graph Theory

A graph is a collection of:

- (a) Rows and columns
- (b) Vertices and edges
- (c) Equations
- (d) Matches

Question 15

CUET PG 2021 | Binary Search Tree

The given numbers are inserted into an empty binary search tree in the given order: 10, 1, 3, 5, 15, 12 and 16. What is the height of the binary search tree?

- (a) 3
- (b) 4
- (c) 5
- (d) 6

Question 16

CUET PG 2021 | Graph Representation

Which of the following ways can be used to represent a graph?

- (a) Only Adjacency list
- (b) Only Adjacency matrix
- (c) Only Incidence matrix
- (d) All of the above

Question 17

CUET PG 2021 | B* Tree

In a B* tree, if the search-key value is 8 bytes long, the block size is 512 bytes and the block pointer size is 2 bytes, then maximum order of the B* tree is:

- (a) 50
- (b) 51
- (c) 52
- (d) 54

Question 18

CUET PG 2021 | Binary Trees Counting

The number of different binary trees with 6 nodes is:

- (a) 6
- (b) 42
- (c) 132
- (d) 256

Question 19

CUET PG 2021 | Greedy Algorithms

Which of the following standard algorithms is not a greedy algorithm?

- (a) Prim's algorithm
- (b) Kruskal algorithm
- (c) Dijkstra's shortest path algorithm
- (d) Bellman-Ford shortest path algorithm

Question 20

CUET PG 2021 | Prim's Algorithm

Prim's algorithm is a/an:

- (a) Divide and conquer algorithm
- (b) Greedy algorithm
- (c) Dynamic programming
- (d) Approximation algorithm

Question 21

CUET PG 2024 | Bubble Sort

The following integers are needed to be sorted in ascending order using bubble sort: 5, 8, 22, 18, 1. Following are the results of various passes:

- (A) 5, 1, 8, 18, 22
- (B) 1, 5, 8, 18, 22
- (C) 5, 8, 18, 1, 22
- (D) 5, 8, 1, 18, 22

- (a) C, D, B, A
- (c) D, C, A, B

- (b) C, D, A, B
- (d) D, C, B, A

Question 22

CUET PG 2024 | Asymptotic Complexities

Arrange the following in the increasing order of their asymptotic complexities:

- (A) Insertion sort (best case)
- (B) Bubble sort (worst case)
- (C) Binary search (worst case)
- (D) Merge sort (worst case)

- (a) (A), (C), (B), (D)
- (c) (A), (B), (C), (D)

- (b) (D), (A), (C), (B)
- (d) (C), (A), (D), (B)

Question 23

CUET PG 2024 | Linked List

Each node is having a successor node in:

- (a) Singly linked list
- (c) Doubly linked list

- (b) Singly circular linked list
- (d) Not possible in any linked list

Question 24

CUET PG 2024 | Linked List Traversal

If we want to find last node of a singly linked list then the correct coding is:

- (a) If (temp->link != NULL) temp = temp->link
- (c) While (temp->link != NULL) temp = temp->link

- (b) If (temp->data == Num) temp = temp->link
- (d) While (temp->link != data) temp = temp->link

Question 25

CUET PG 2024 | Priority Queue

What are the ways to implement a priority Queue?

- (A) Arrays
- (B) Fibonacci tree
- (C) Heap Data Structure
- (D) Linked list

- (a) A, B and D only
- (c) A, B, C and D

- (b) B, C and D only
- (d) A, C and D only

Question 26

CUET PG 2024 | Stack Applications

Which of the following is not an application of Stack?

- (a) Tower of Hanoi
- (c) Voter polling station

- (b) Recursion
- (d) Parentheses matching

Question 27

CUET PG 2024 | B+ Tree

Consider implementation of a database. Among the following options, choose the most appropriate data structure for this:

- (a) B+ tree
- (b) Linked list
- (c) Queue
- (d) Stack

Question 28

CUET PG 2024 | BST Worst Case

Consider a completely skewed (left/right) binary search tree with n elements. What is the worst case time complexity of searching an element in this tree?

- (a) $O(n)$
- (b) $O(1)$
- (c) $O(\log n)$
- (d) $O(n \log n)$

Question 29

CUET PG 2025 | Binary Tree Identification

Consider the following tree (nodes: 105, 165, 125). This is a/an:

- (a) AVL Search Tree
- (b) Binary Tree
- (c) Binary Search Tree
- (d) Fibonacci Tree

Question 30

CUET PG 2025 | Graph Theory BFS vs DFS

Consider the task of finding the shortest path in an unweighted graph by using BFS and DFS. Which of the following statements are true?

- (A) BFS always finds the shortest path.
 - (B) DFS always finds the shortest path.
 - (C) DFS does not guarantee finding the shortest path.
 - (D) BFS does not guarantee finding the shortest path.
- (a) (B) and (D) only
 - (b) (A) and (C) only
 - (c) (A) and (B) only
 - (d) (C) and (D) only

Question 31

CUET PG 2025 | DFS Application

Which of the following is not an application of DFS?

- (a) Topological sorting
- (b) Determining Strongly Connected Components in a graph
- (c) Finding minimum distance to a node in an unweighted graph optimally
- (d) Solving Maze Problem

Question 32

CUET PG 2025 | Recurrence Relations

Match List-I with List-II:

List-I (Algorithm): (A) Binary Search, (B) Merge Sort, (C) Quick Sort (worst case), (D) Linear Search

List-II (Recurrence relation): (I) $T(n) = T(n/2) + c$, (II) $T(n) = 2T(n/2) + \Theta(n)$, (III) $T(n) = T(n-1) + \Theta(n)$, (IV) $T(n) = T(n-1) + c$

- (a) (A)-(I), (B)-(II), (C)-(III), (D)-(IV)
- (b) (A)-(I), (B)-(III), (C)-(II), (D)-(IV)
- (c) (A)-(I), (B)-(II), (C)-(IV), (D)-(III)
- (d) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)

Question 33

CUET PG 2025 | Quicksort

The Quicksort and randomized Quicksort procedures differ in:

- (a) Selection of Pivot element
- (b) Worst case time complexity
- (c) Best case time complexity
- (d) Final Output

Question 34

CUET PG 2025 | Heap Properties

Given the index i of a node in a heap, we cannot compute:

- (a) Parent (i)
- (b) Heap size
- (c) Left (i)
- (d) Right (i)

Question 35

CUET PG 2025 | BST Time Complexity

In a binary search tree, the worst case time complexity of inserting and deleting a key is:

- (a) $O(\log n)$ for insertion and $O(\log n)$ for deletion
- (b) $O(n)$ for insertion and $O(\log n)$ for deletion
- (c) $O(n)$ for insertion and $O(n)$ for deletion
- (d) $O(\log n)$ for insertion and $O(n)$ for deletion

Question 36

CUET PG 2025 | Time Complexity

Arrange the following time complexities in increasing order:

- (A) Bubble sort (worst case)
- (B) Deleting head node in singly linked list
- (C) Binary search
- (D) Worst case of merge sort

- (a) (A), (B), (C), (D)
- (b) (B), (C), (D), (A)
- (c) (B), (A), (D), (C)
- (d) (C), (B), (D), (A)

Question 37

CUET PG 2025 | BST Operations

In case of Binary search tree which of the following procedure's running time is distinct among all?

- (a) TREE-SUCCESSOR (finds successor of the given node)
- (b) TREE-MAXIMUM (finds the node with maximum value)
- (c) INORDER-WALK (prints all elements of a tree in inorder manner)
- (d) TREE-MINIMUM (finds the node with minimum value)

Question 38

CUET PG 2026 | Graph Representation & Complexity

Which of the following statements are correct?

- A. In a directed graph, the sum of lengths of all the adjacency list is $|E|$.
- B. The adjacency matrix requires $\Theta(V^2)$ memory.
- C. In an undirected graph, the sum of lengths of all the adjacency list is $|E|$.
- D. The memory requirement of adjacency list is dependent on the number of edges in the graph.

- (1) A, B, C only
- (2) B, C, D only
- (3) A, B, D only
- (4) A, C, D only

Question 39

CUET PG 2026 | Trees

A tree used to represent arithmetic expressions is called

- | | |
|--------------------|---------------------|
| (1) Threaded tree | (2) Game tree |
| (3) Red-Black tree | (4) Expression tree |

Question 40

CUET PG 2026 | Data Structures Match List

Match List - I with List II.

List - I: A. Breadth first search, B. Sparse Graph, C. Collision Resolution, D. Linked list

List - II: I. Shortest paths, II. Adjacency list representation, III. Open addressing, IV. Self referential structures

- | | |
|----------------------------|----------------------------|
| (1) A-I, B-II, C-III, D-IV | (2) A-I, B-II, C-IV, D-III |
| (3) A-II, B-I, C-III, D-IV | (4) A-IV, B-III, C-II, D-I |

Question 41

CUET PG 2026 | Linked List Search Algorithm

Consider a procedure LIST-SEARCH (L, K) finds the first elements with key K in list L by simple linear search, returning a pointer to its element. Arrange the following according to correct order of the execution.

A. $x = L.head$

B. While $x \neq NIL$ and $x.key \neq K$

C. return x

D. $x = x.next$

- | | |
|----------------|----------------|
| (1) A, B, D, C | (2) B, C, D, A |
| (3) A, C, B, D | (4) A, B, C, D |

Question 42

CUET PG 2026 | Graph Search

In a search, we begin search at a starting node, and when we come across a dead end then we backtrack.

- | | |
|-------------------|---------------|
| (1) Breadth first | (2) Connected |
| (3) Depth first | (4) Bubble |

Question 43

CUET PG 2026 | Arrays Indexing

The length or the number of data elements of the array can be obtained from the index set by which of the following formula?

- | | |
|----------------------------------|---------------------------------|
| (1) Lower Bound - Upperbound - 1 | (2) Upperbound - Lowerbound + 1 |
| (3) Lowerbound - Upperbound + 1 | (4) Upperbound + Lowerbound - 1 |

Question 44

CUET PG 2026 | Linked Lists

In a linked list, the last node's link field points to the first node of the list.

- | | |
|----------------|-------------|
| (1) Circular | (2) Linear |
| (3) Sequential | (4) Indexed |

Question 45

CUET PG 2026 | Stack Applications

Which of the following is NOT an application of stack?

- | | |
|---------------------|--------------------------------------|
| (1) Towers of Hanoi | (2) Evaluation of Postfix expression |
| (3) Recursion | (4) Priority Queue |

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ANSWER KEY (2021 – 2026)

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	a	2	d	3	b	4	b	5	b
6	b	7	c	8	b	9	b	10	b
11	c	12	a	13	d	14	b	15	a
16	d	17	c	18	b	19	d	20	b
21	c	22	d	23	b	24	c	25	d
26	c	27	a	28	a	29	b	30	b
31	c	32	c	33	a	34	b	35	c
36	b	37	c	38	4	39	4	40	3
41	1	42	3	43	2	44	1	45	4

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